MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RELAY, ANALOG, KEY CODE RRF

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic relay module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

Name	Relay
Family	Analog
Key code	RRF
Size	1B
Weight	59 grams maximum
Environment	Class I or class II as specified in MIL-STD-1389

	Class I	Class II	Unit
Failure rate	0.093 max	0.138 max	(failures/10^6 hours)
Maximum power dissipation:			
32 Vdc. - 55°C		5.0 max	W
26 Vdc. 25°C	2.28	2.28	W
Coil voltage and current per relay	32 Vdc at 15.8 mA (max)	32 Vdc at 15.8 mA (max)	
Contact rating	28 Vdc at 1.0 A (max)	28 Vdc at 1.0 A (max)	
Pickup voltage	6.0 to 20.0 max	6.0 to 20.0 max	Vdc
Dropout voltage	0.89 to 13.0 max	0.89 to 13.0 max	Vdc
Operate time	4.0 max	4.0 max	ms
Release time	7.5 max	7.5 max	ms
Contact bounce (pickup and dropout)	1.5 max	1.5 max	ms

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Module function. This module contains 10 relays, each with two normally-open and two normally-closed contacts. Across each relay coil is a diode for suppression of switching transients.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

MIL-R-39016/16 - Relay, Electromagnetic, Established Reliability, DPDT, Low Level to 1.0 Ampere (Sevsitive, 6.0 Milliwatts) with Internal Diode for Coil Transient Suppression.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Module Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.093 failures per million operating hours for class I or 0.138 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II. Class I modules shall conform the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RRF.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 59 grams.

3.5.4 Contact pins. The module shall have 80 contact pins positioned A1 through A40 and B1 through B40.

3.5.5 Fin position. The module shall have an insertion and extraction fin on increment A and heat dissipation fins on increments A and B.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/279-1 for class I and M28787/279-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

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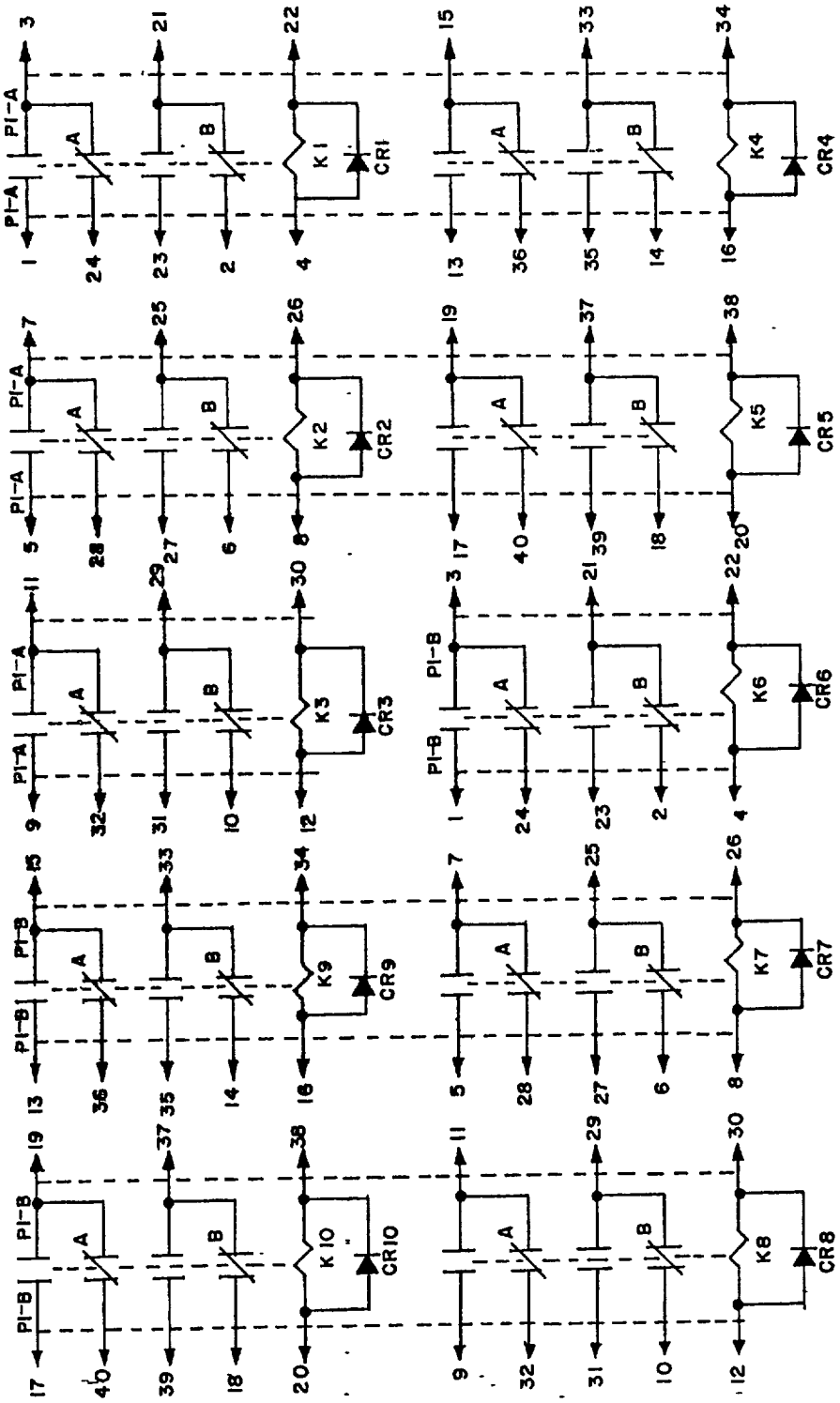


FIGURE 1. RRF module function diagram, relay.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	25° ±5°C (see note 4)		Class I (see note 4)				Class II (see note 4)				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A45	Module integrity													
	Initial	II	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II		1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B40	Open contact resistance													
	Initial	III	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III		1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
C1-C40	Closed contact resistance check													
	Initial	IV	X	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	Ω
	End-of-life	IV		0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	Ω
D1-D10	Diode forward resistance check	V	X	10	100	10	100	10	100	10	100	10	100	Ω
E1-E10	Diode inverse resistance check	VI	X	2.97	3.63	2.67	4.50	2.67	4.50	2.02	4.72	2.02	4.72	kΩ
F1-F10	Pickup voltage	VII	X	8.0	14.0	7.2	18.0	7.2	18.0	6.0	20.0	6.0	20.0	V
G1-G10	Dropout voltage	VIII	X	1.4	8.0	1.2	9.0	1.2	9.0	0.89	13.0	0.89	13.0	V
H1-H20	Operate time	IX	S	1.0	4.0	1.0	4.0	1.0	4.0	1.0	4.0	1.0	4.0	ms
J1-J20	Release time	X	S	1.0	7.5	1.0	7.5	1.0	7.5	1.0	7.5	1.0	7.5	ms
K1-K10 K21-K30	Contact bounce (on pickup)	XI	Q	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	ms
K11-K20 K31-K40	Contact bounce (on dropout)	XI	Q	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	ms

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or the modules from each inspection lot, whichever is greater, at 25°C.
3. (Q) parameters shall be tested during qualification testing only.
4. Unless otherwise specified, limits are for initial and end-of-life.

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3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I

3.9 Relay requirements. Relays used in this module shall be MIL-R-39016/16-004M or approved equivalent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	2
Square wave generator	Item H01	1
Electronic load	Item EL01	1
DC power supply	Item P06	1
Megohm bridge	Item R06	1
Ohmmeter	Item R10	1
Digital multifunction meter	Item M02	1
DC ammeter	Item X05	1
DC power supply	Item P07	2
Oscilloscope	Item D01	1
Oscilloscope probe	Item Y01	2
Resistors		
499Ω ±1%, 1/10 W		1
3.0Ω ±1%, 1/10 W		10
14.0Ω ±1%, 20 W, noninductive		10
120 Ω1%, 1 W		1
49.9Ω ±1%, 1/8 W		1
28kΩ ±1%, 1/4 W		1
Transistor		
2N3108 or equivalent		1

All resistors shall be metal film unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through K shall be performed in any order with the following exceptions: If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through K in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through K shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature tests. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the test conditions A through G in table I shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Coil life test. The coil life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the coil life test and while still at the maximum class temperature, electrical tests B through K shall be performed in the order specified in 4.4.2 using the maximum class temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

4.5.4 Contact life test. The contact life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. The module shall be run for 100,000 cycles minimum. Following the contact life test and while still at the maximum class temperature, electrical tests B through K shall be performed in the order specified in 4.4.2 using the maximum class temperature end-of-life requirements of table I. Following the return of the module temperature to 25°C, electrical tests A through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C end-of-life requirement of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Preparing activity.
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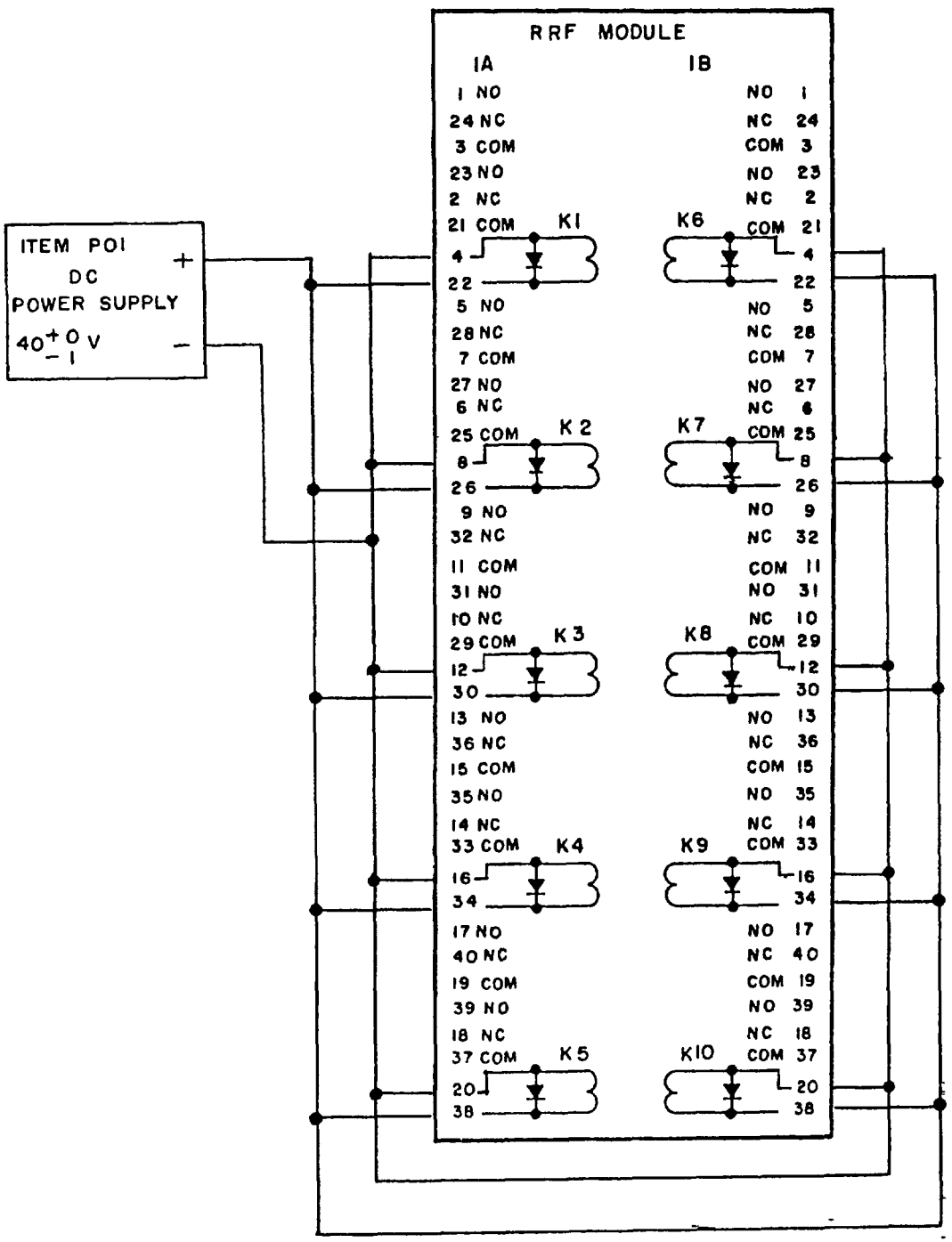
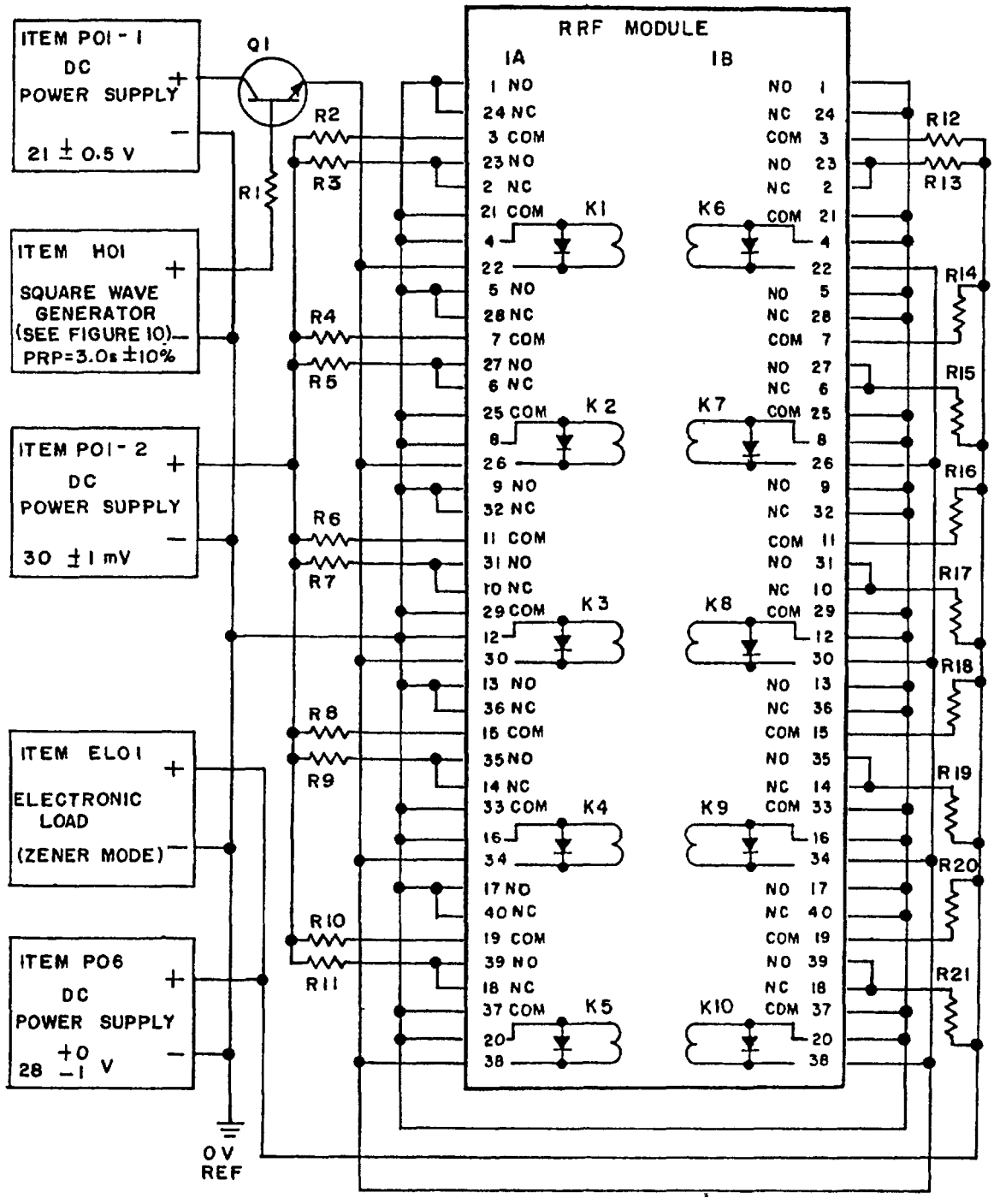


FIGURE 2. Operating temperature, component temperature and coil life test circuit.

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NOTES:
1. R1 = 499Ω ±1%, 1/10 W.
2. R2-R11 = 3Ω ±1%, 1/10 W.
3. R12-R21 = 14Ω ±1%, 20 W non-inductive.
4. Q1 = 2N3108 or equivalent.

FIGURE 3. Contact life-test circuit.

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TABLE II. Module integrity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Module integrity	1A1,1A3,1A21,1A23,	1A4, 1A22	10.0	∞	MΩ	4
A2		1A13,1A15,1A33,1A35	1A16, 1A34				
A3		1A5,1A7,1A25,1A27	1A8, 1A26				
A4		1A17,1A19,1A37,1A39	1A20, 1A38				
A5		1A9,1A11,1A29,1A31	1A12, 1A30				
A6		1A1,1A3	1A21,1A23				
A7		1A13,1A15	1A33,1A35				
A8		1A5,1A7	1A25,1A27				
A9		1A17,1A19	1A37,1A39				
A10		1A9,1A11	1A29,1A31				
A11		1A1-1A4,1A21-1A24	1A13-1A16,1A33-1A36				
A12		1A1-1A4,1A21-1A24	1A5-1A8,1A25-1A28				
A13		1A1-1A4,1A21-1A24	1A17-1A20,1A37-1A40				
A14		1A1-1A4,1A21-1A24	1A9-1A12,1A29-1A32				
A15		1A13-1A16,1A33-1A36	1A5-1A8,1A25-1A28				
A16		1A13-1A16,1A33-1A36	1A17-1A20,1A37-1A40				
A17		1A13-1A16,1A33-1A36	1A9-1A12,1A29-1A32				
A18		1A5-1A8,1A25-1A28	1A17-1A20,1A37-1A40				
A19		1A5-1A8,1A25-1A28	1A9-1A12,1A29-1A32				
A20	Module integrity	1A17-1A20,1A37-1A40	1A9-1A12,1A29-1A32	10.0	∞	MΩ	4

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TABLE II. Module integrity test. - Continued

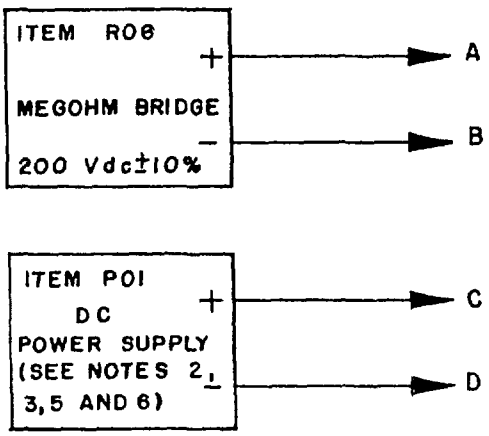
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A21	Module integrity (open contact)	1A3,1A7,1A11,1A15,1A19,1A21 1A25,1A29,1A33,1A37	1A1,1A5,1A9,1A13,1A17,1A23, 1A27,1A31,1A35,1A39	10.0	-	MΩ	4
A22	Module integrity (open contact)	1A3,1A7,1A11,1A15,1A19,1A21, 1A25,1A29,1A33,1A37	1A2,1A6,1A10,1A14,1A18,1A24 1A28,1A32,1A36,1A40				
A23	Module integrity	1B1,1B3,1B21,1B23	1B4				
A24		1B13,1B15,1B33,1B35	1B16				
A25		1B5,1B7,1B25,1B27	1B8				
A26		1B17,1B19,1B37,1B39	1B20				
A27		1B9,1B11,1B29,1B31	1B12				
A28		1B1,1B3	1B21,1B23				
A29		1B13,1B15	1B33,1B35				
A30		1B5,1B7	1B25,1B27				
A31		1B17,1B19	1B37,1B39				
A32		1B9,1B11	1B29,1B31				
A33		1B1-1B4,1B21-1B24	1B13-1B16,1B33-1B36				
A34		1B1-1B4,1B21-1B24	1B5-1B8,1B25-1B28				
A35		1B1-1B4,1B21-1B24	1B17-1B20,1B37-1B40				
A36		1B1-1B4,1B21-1B24	1B9-1B12,1B29-1B32				
A37		1B13-1B16,1B33-1B36	1B5-1B8,1B25-1B28				
A38		1B13-1B16,1B33-1B36	1B17-1B20,1B37-1B40				
A39		1B13-1B16,1B33-1B36	1B9-1B12,1B29-1B32				
A40	Module integrity	1B5-1B8,1B25-1B28	1B17-1B20,1B37-1B40	10.0	-	MΩ	4

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TABLE II. Module integrity test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A41	Module integrity	1B5-1B8,1B25-1B28	1B9-1B12,1B29-1B32	10.0	∞	MΩ	4
A42	Module integrity	1B17-1B20,1B37-1B40	1B9-1B12,1B29-1B32	↓	↓	↓	↓
A43	Module integrity (open contact)	1B3,1B7,1B11,1B15,1B19,1B21, 1B25,1B29,1B33,1B37	1B1,1B5,1B9,1B13,1B17,1B23, 1B27,1B31,1B35,1B39				
A44	Module integrity (open contact)	1B3,1B7,1B11,1B15,1B19,1B21, 1B25,1B29,1B33,1B37	1B2,1B6,1B10,1B14,1B18,1B24 1B28,1B32,1B36,1B40				
A45	Module integrity (open contact)	1A1 thru 1A40	1B1 thru 1B40	10.0	∞	MΩ	4

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. Only during test conditions A21 and A22 connect C to pins 1A22, 1A26, 1A30, 1A34, and 1A38; connect D to pins 1A4, 1A8, 1A12, 1A16, and 1A20.
2. For test condition A21 adjust item P01 to 0.88 ±0.1 Vdc.
3. For test condition A22 adjust item P01 to 20.0 ±0.7 Vdc.
4. Only during test conditions A43 and A44 connect C to pins 1B22, 1B26, 1B30, 1B34, and 1B38; connect D to pins 1B4, 1B8, 1B12, 1B16, and 1B20.
5. For test condition A43 adjust item P01 to 0.88 ±0.1 Vdc.
6. For test condition A44 adjust item P01 to 20.0 ±0.7 Vdc.

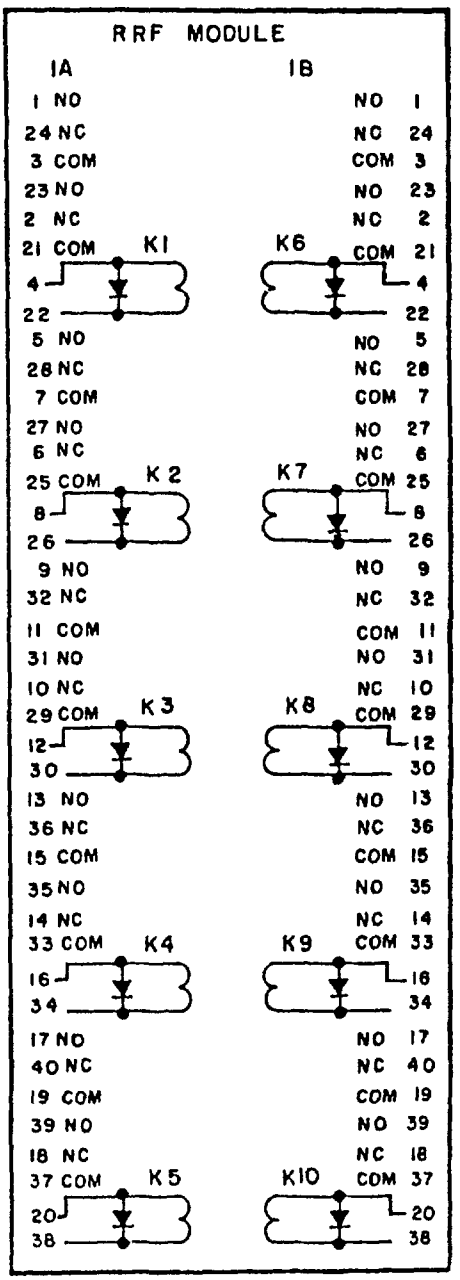


FIGURE 4. Module integrity test circuit for table II.

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TABLE III. Open contact resistance test.

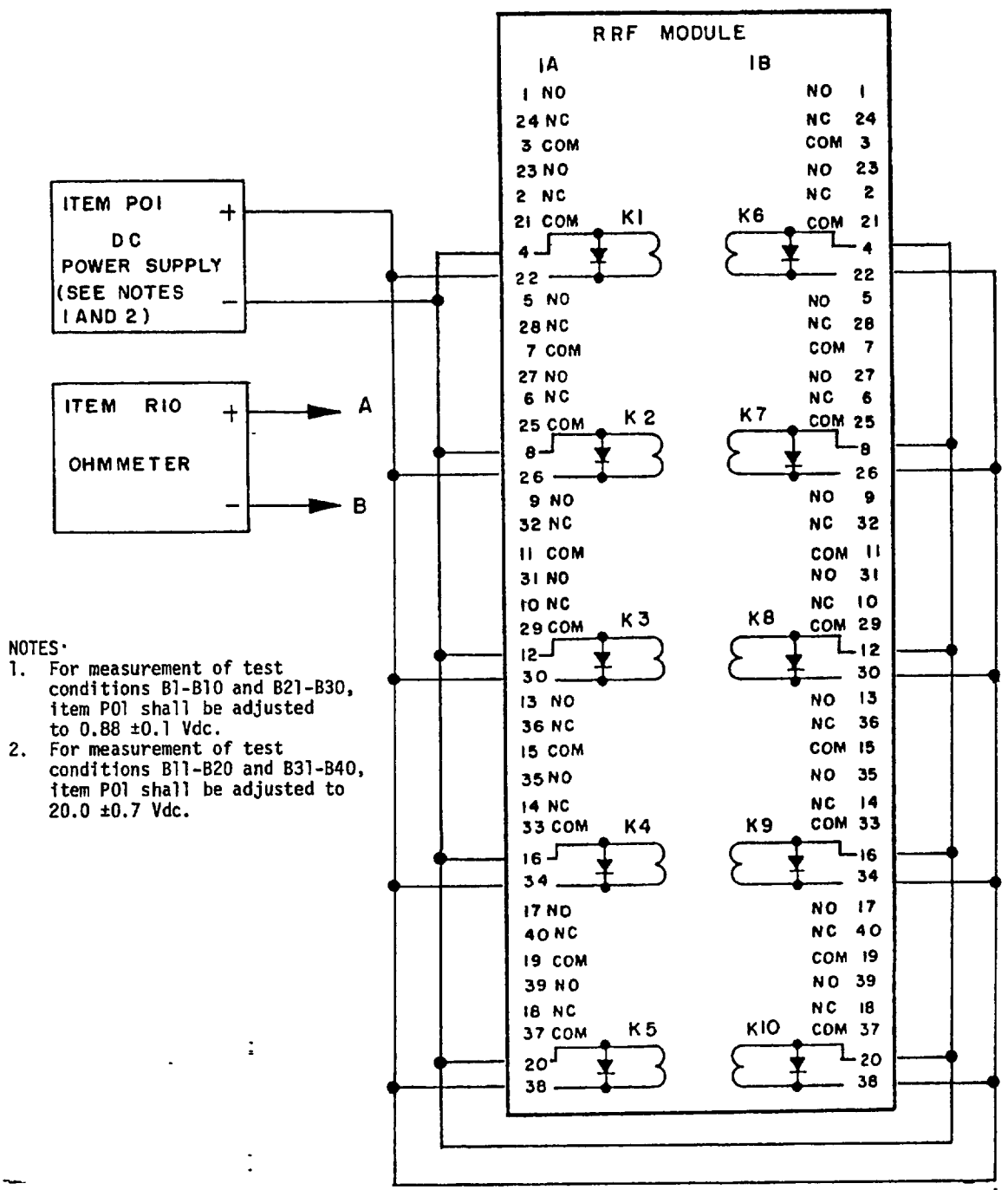
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Open contact resistance	1A3	1A1	10.0	∞	MΩ	5
B2		1A21	1A23				
B3		1A7	1A5				
B4		1A25	1A27				
B5		1A11	1A9				
B6		1A29	1A31				
B7		1A15	1A13				
B8		1A33	1A35				
B9		1A19	1A17				
B10		1A37	1A39				
B11		1A3	1A24				
B12		1A21	1A2				
B13		1A7	1A28				
B14		1A25	1A6				
B15		1A11	1A32				
B16		1A29	1A10				
B17		1A15	1A36				
B18		1A33	1A14				
B19		1A19	1A40				
B20	Open contact resistance	1A37	1A18	10.0	∞	MΩ	5

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TABLE III. Open contact resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B21	Open contact resistance	1B3	1B1	10.0	∞	MΩ	5
B22		1B21	1B23				
B23		1B7	1B5				
B24		1B25	1B27				
B25		1B11	1B9				
B26		1B29	1B31				
B27		1B15	1B13				
B28		1B33	1B35				
B29		1B19	1B17				
B30		1B37	1B39				
B31		1B3	1B24				
B32		1B21	1B2				
B33		1B7	1B28				
B34		1B25	1B6				
B35		1B11	1B32				
B36		1B29	1B10				
B37		1B15	1B36				
B38		1B33	1B14				
B39		1B19	1B40				
B40	Open contact resistance	1B37	1B18	10.0	∞	MΩ	5

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NOTES:
1. For measurement of test conditions B1-B10 and B21-B30, item P01 shall be adjusted to 0.88 ±0.1 Vdc.
2. For measurement of test conditions B11-B20 and B31-B40, item P01 shall be adjusted to 20.0 ±0.7 Vdc.

FIGURE 5. Open contact resistance test circuit for table III.

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TABLE IV. Closed contact resistance check test. 1/

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Closed contact resistance check	1A3	1A24	0.0	2.0	mV	6
C2		1A21	1A2				
C3		1A15	1A36				
C4		1A33	1A14				
C5		1A7	1A28				
C6		1A25	1A6				
C7		1A19	1A40				
C8		1A37	1A18				
C9		1A11	1A32				
C10		1A29	1A10				
C11		1A3	1A1				
C12		1A21	1A23				
C13		1A15	1A13				
C14		1A33	1A35				
C15		1A7	1A5				
C16		1A25	1A27				
C17		1A19	1A17				
C18		1A37	1A39				
C19	Closed contact resistance check	1A11	1A9	0.0	2.0	mV	6

See footnote at end of table.

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TABLE IV. Closed contact resistance check test 1/ - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C20	Closed contact resistance check	1A29	1A31	0.0	2.0	mV	6
C21		1B3	1B24				
C22		1B21	1B2				
C23		1B15	1B36				
C24		1B33	1B14				
C25		1B7	1B28				
C26		1B25	1B6				
C27		1B19	1B40				
C28		1B37	1B18				
C29		1B11	1B32				
C30		1B29	1B10				
C31		1B3	1B1				
C32		1B21	1B23				
C33		1B15	1B13				
C34		1B33	1B35				
C35		1B7	1B5				
C36		1B25	1B27				
C37	Closed contact resistance check	1B19	1B17	0.0	2.0	mV	6

See footnote at end of table.

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TABLE IV. Closed contact resistance check test 1/ - Continued.

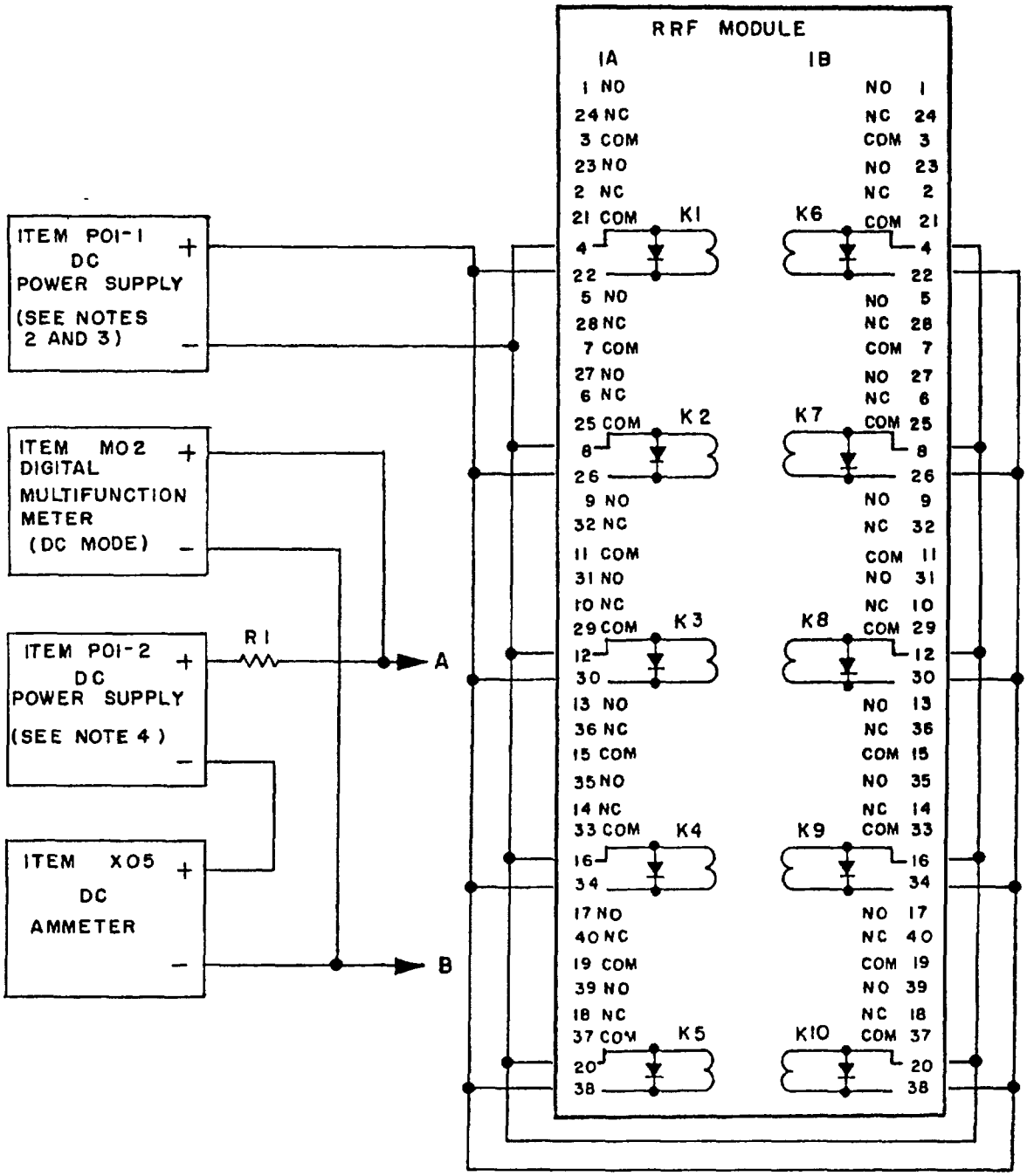
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C38	Closed contact resistance check	1B37	1B39	0.0	2.0	mV	6
C39	Closed contact resistance check	1B11	1B9	0.0	2.0	mV	6
C40	Closed contact resistance check	1B29	1B31	0.0	2.0	mV	6

1/ The closed contact resistance in ohms can be determined from the following equation:

$$R = \frac{V}{I} = \frac{V}{10}$$

where, V = the measured voltage across the relay contact in mV,
and I = the current through the relay of 10 mA.

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NOTES:
1. R1 = 120Ω ±1%, 1 W.
2. For test conditions C1-C10 and C21-C30 adjust item P01-1 to 0.88 ±0.1 Vdc..
3. For test conditions C11-C20 and C31-C40 adjust item P01-1 to 20.0 ±0.7 Vdc.
4. For test conditions C1-C40 adjust item P01-2 for a current of 10 mA ±1% as measured on item X05.
5. Voltage sensing leads from the positive and negative terminals of item M02 to the module pins at points A and B shall be connected in such a way as to exclude the resistance (or voltage drop at 10 mA) of the current-carrying leads.

FIGURE 6. Closed contact resistance check test circuit for table IV.

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TABLE V. Diode forward resistance check test. 1/

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Diode forward resistance check	1A4	Open	1A22	A	0.10	1.00	V	7
D2		1A16		1A34					
D3		1A8		1A26					
D4		1A20		1A38					
D5		1A12		1A30					
D6		1B4		1B22					
D7		1B16		1B34					
D8		1B8		1B26					
D9		1B20		1B38					
D10	Diode forward resistance check	1B12	Open	1B30	A	0.10	1.00	V	7

1/ $\frac{\text{T.C.D (Volts)}}{\text{0.5 V/49.9}\Omega}$ = table I limits (10 to 100Ω)

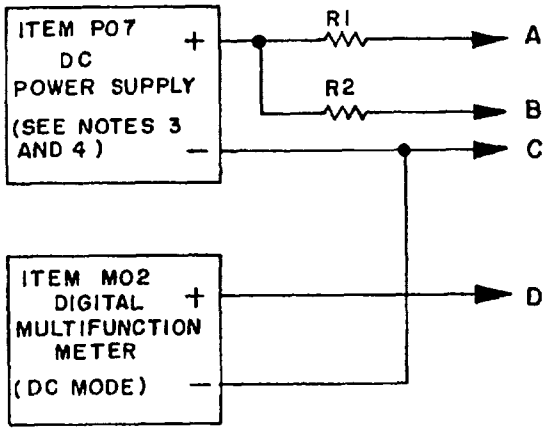
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TABLE VI. Diode inverse resistance check test. 1/

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	Diode inverse resistance check	Open	1A22	1A4	B	2.97	3.63	V	7
E2			1A34	1A16					
E3			1A26	1A8					
E4			1A38	1A20					
E5			1A30	1A12					
E6			1B22	1B4					
E7			1B34	1B16					
E8			1B26	1B8					
E9			1B38	1B20					
E10	Diode inverse resistance check	Open	1B30	1B12	B	2.97	3.63	V	7

1/ $\frac{\text{T.C.E (Volts)}}{\text{0.5 V/499}\Omega}$ = table I limits (2.97 kΩ to 3.63 kΩ)

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NOTES:
1. R1 = 49.9Ω ±1%, 1/8 W.
2. R2 = 499Ω ±1%, 1/10 W.
3. After connecting points A, B, and C adjust item P07 for a drop of 0.5 V ±1% across R1 before taking reading for test conditions D1 through D10.
4. After connecting points A, B, and C, adjust item P07 for a drop of 0.5 V ±1% across R2 before taking readings for test conditions E1 through E10.

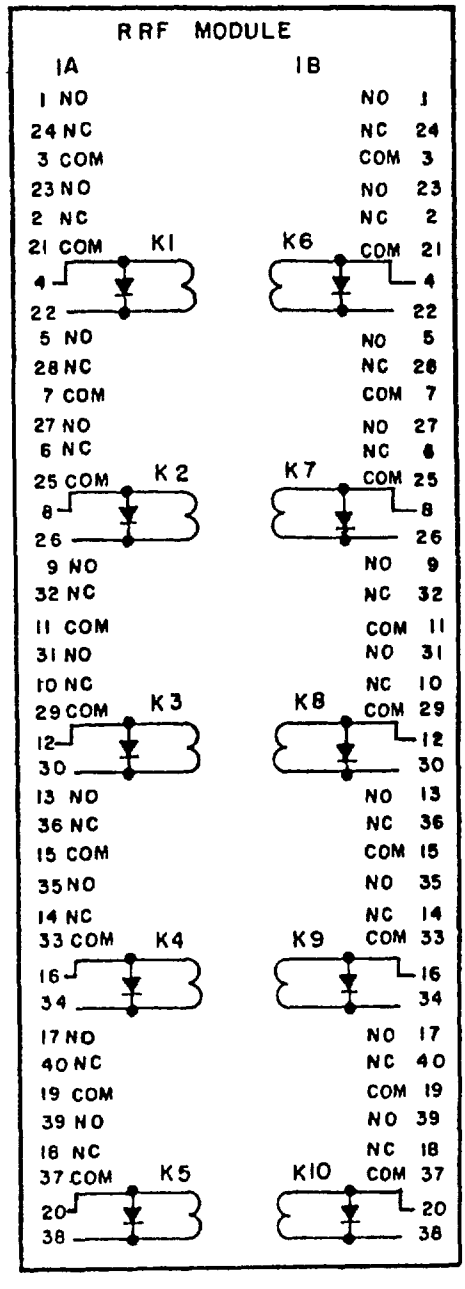


FIGURE 7. Diode forward resistance and diode inverse resistance check test circuit for tables V and VI.

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TABLE VII. Pickup voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	Pickup voltage	1A22	1A4	1A3	1A1	8.0	14.0	V	8
F2	↓	1A34	1A16	1A15	1A13	↓	↓	↓	↓
F3		1A26	1A8	1A7	1A5				
F4		1A38	1A20	1A19	1A17				
F5		1A30	1A12	1A11	1A9				
F6		1B22	1B4	1B3	1B1				
F7		1B34	1B16	1B15	1B13				
F8		1B26	1B8	1B7	1B5				
F9		1B38	1B20	1B19	1B17				
F10	Pickup voltage	1B30	1B12	1B11	1B9	8.0	14.0	V	8

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TABLE VIII. Dropout voltage test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	Dropout voltage	1A22	1A4	1A3	1A24	1.4	8.0	V	8
G2		1A34	1A16	1A15	1A36				
G3		1A26	1A8	1A7	1A28				
G4		1A38	1A20	1A19	1A40				
G5		1A30	1A12	1A11	1A32				
G6		1B22	1B4	1B3	1B24				
G7		1B34	1B16	1B15	1B36				
G8		1B26	1B8	1B7	1B28				
G9		1B38	1B20	1B19	1B40				
G10	Dropout voltage	1B30	1B12	1B11	1B32	1.4	8.0	V	8

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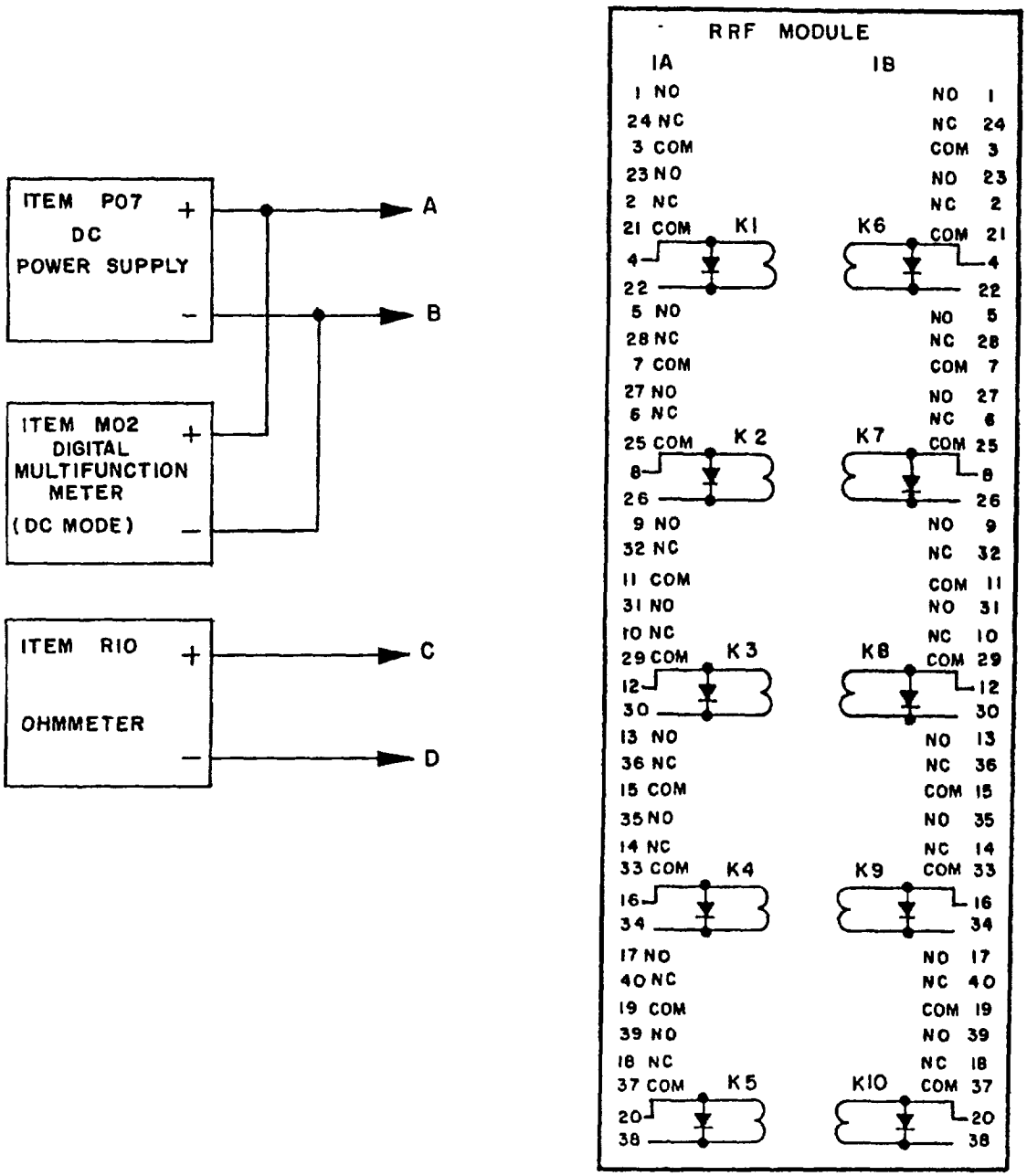


FIGURE 8. Pick-up voltage and drop-out voltage test circuit for tables VII and VIII.

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NOTES:
1. Test procedure for test conditions F1-F10.

   a. Connect test points A, B, C, and D as specified for test condition F1.
   b. Adjust output of item P07 for 31 ±1 V as measured on item M02. Maintain this voltage for a period of 1 to 3 seconds.
   c. Gradually reduce the output voltage of item P07 to zero. The contact resistance, as measured on item R10, should be 1 megohm or greater.
   d. Gradually increase the output voltage of item P07 until the contact resistance, as measured on item R10, drops to 1 ohm or less. However, do not exceed 40 V.
   e. The pick-up voltage shall be read on item M02, and shall be as specified for test condition F1.
   f. Repeat notes a through e for test conditions F2-F10.

2. Test procedure for test conditions G1-G10.

   a. Connect test points A, B, C, and D as specified for test condition G1.
   b. Adjust output of item P07 for 31 ±1 V as measured on item M02. The contact resistance, as measured on item R10 should be 1 megohm or greater.
   c. Gradually reduce the output voltage of item P07 until the contact resistance, as measured on item R10 drops to 1 ohm or less.
   d. The drop-out voltage shall be read on item M02 and shall be as specified for test condition G1.
   e. Repeat notes a through d for test conditions G2-G10.

FIGURE 8. Pick-up voltage and drop-out voltage test circuit for tables VII and VIII. - Continued

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TABLE IX. Operate time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Operate time	1A3	1A1	1.0	4.0	ms	9,10
H2		1A21	1A23				
H3		1A15	1A13				
H4		1A33	1A35				
H5		1A7	1A5				
H6		1A25	1A27				
H7		1A19	1A17				
H8		1A37	1A39				
H9		1A11	1A9				
H10		1A29	1A31				
H11		1B3	1B1				
H12		1B21	1B23				
H13		1B15	1B13				
H14		1B33	1B35				
H15		1B7	1B5				
H16		1B25	1B27				
H17		1B19	1B17				
H18		1B37	1B39				
H19		1B11	1B9				
H20	Operate time	1B29	1B31	1.0	4.0	ms	9,10

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TABLE X. Release time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Release time	1A3	1A24	1.0	7.5	ms	9,10
J2		1A21	1A2				
J3		1A15	1A36				
J4		1A33	1A14				
J5		1A7	1A28				
J6		1A25	1A6				
J7		1A19	1A40				
J8		1A37	1A18				
J9		1A11	1A32				
J10		1A29	1A10				
J11		1B3	1B24				
J12		1B21	1B2				
J13		1B15	1B36				
J14		1B33	1B14				
J15		1B7	1B28				
J16		1B25	1B6				
J17		1B19	1B40				
J18		1B37	1B18				
J19		1B11	1B32				
J20	Release time	1B29	1B10	1.0	7.5	ms	9,10

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TABLE XI. Contact time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Contact bounce (on pickup)	1A3	1A1	0.0	1.5	ms	9,10
K2	↓	1A21	1A23	↓	↓	↓	↓
K3		1A15	1A13				
K4		1A33	1A35				
K5		1A7	1A5				
K6		1A25	1A27				
K7		1A19	1A17				
K8		1A37	1A39				
K9		1A11	1A9				
K10	Contact bounce (on pickup)	1A29	1A31				
K11	Contact bounce (on dropout)	1A3	1A24				
K12	↓	1A21	1A2				
K13		1A15	1A36				
K14		1A33	1A14				
K15		1A7	1A28				
K16		1A25	1A6				
K17		1A19	1A40				
K18		1A37	1A18				
K19		1A11	1A32				
K20	Contact bounce (on dropout)	1A29	1A10	0.0	1.5	ms	9,10

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TABLE XI. Contact bounce test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K21	Contact bounce (on pickup)	1B3	1B1	0.0	1.5	ms	9,10
K22		1B21	1B23				
K23		1B15	1B13				
K24		1B33	1B35				
K25		1B7	1B5				
K26		1B25	1B27				
K27		1B19	1B17				
K28		1B37	1B39				
K29		1B11	1B9				
K30	Contact bounce (on pickup)	1B29	1B31				
K31	Contact bounce (on dropout)	1B3	1B24				
K32		1B21	1B2				
K33		1B15	1B36				
K34		1B33	1B14				
K35		1B7	1B28				
K36		1B25	1B6				
K37		1B19	1B40				
K38		1B37	1B18				
K39		1B11	1B32				
K40	Contact bounce (on dropout)	1B29	1B10	0.0	1.5	ms	9,10

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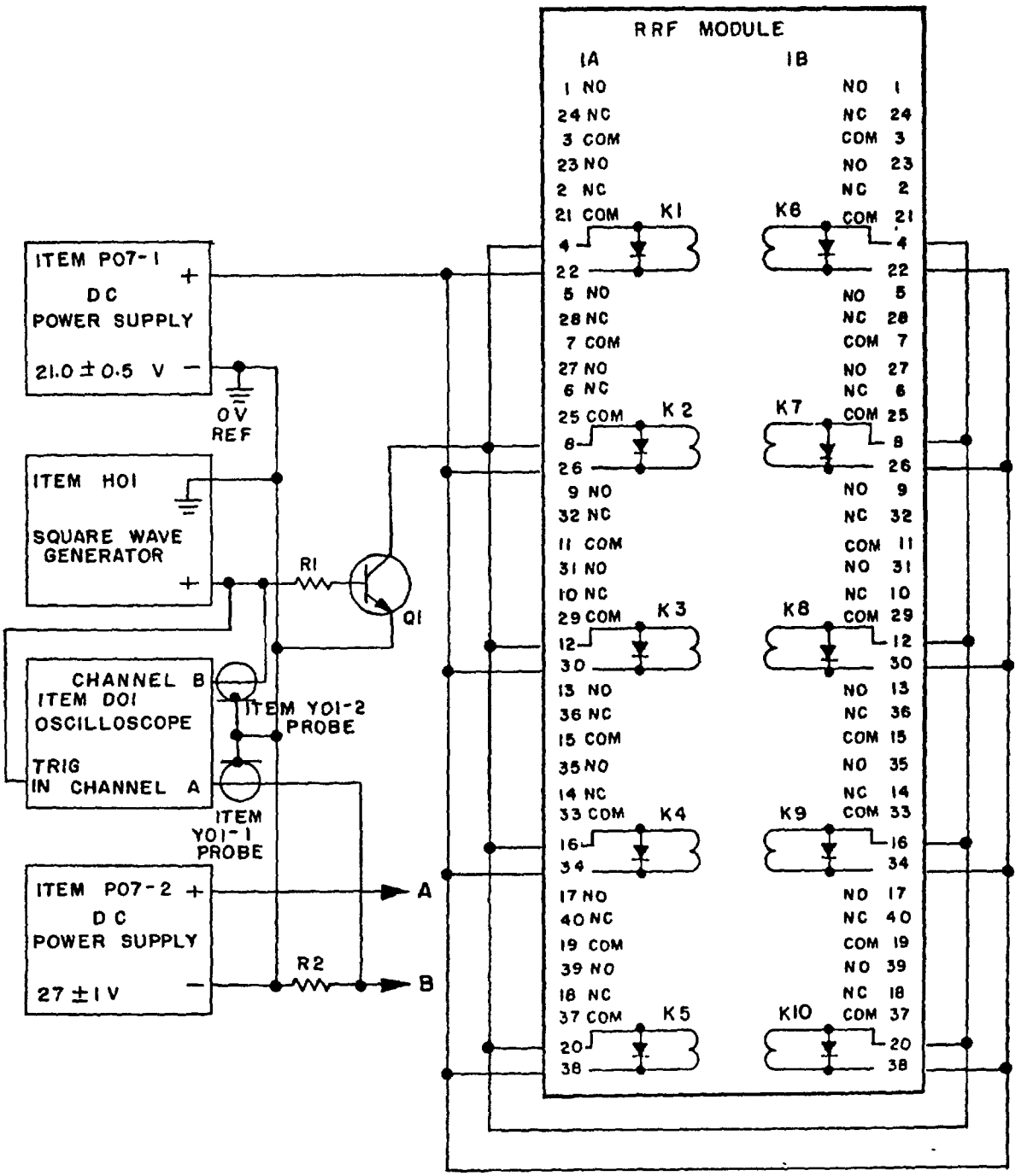


FIGURE 9. Operate time, release time, and contact bounce test circuit for tables IX, X, and XI.

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NOTES:
1. R1 = 499Ω ±1%, 1/10 W.
2. R2 = 28 kΩ ±1%, 1/4 W.
3. Q1 = 2N3108, or equivalent.
4. Test procedure for test conditions H1-H20.

   a. Connect test points A and B as specified for test condition H1.
   b. On item D01, set the trigger coupling to dc, the trigger source to external, and the trigger level and slope to positive.
   c. Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 2.5 ±0.1 Vp as read on item D01, channel B.
   d. The operate time shall be read on item D01, channel A, and shall be as specified for test condition H1.
   e. Repeat notes a through d for test conditions H2-H20.

5. Test procedure for test conditions J1-J20.

   a. Connect test points A and B as specified for test condition J1.
   b. On item D01, set the trigger coupling to dc, the trigger source to external, the trigger level to positive, and the trigger slope to negative.
   c. Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 2.5 ±0.1 Vp as read on item D01, channel B.
   d. The release time shall be read on item D01, channel A, and shall be as specified for test condition J1.
   e. Repeat notes a through d for test conditions J2-J20.

6. Test procedure for test conditions K1-K40.

   a. Connect test points A and B as specified for test condition K1.
   b. On item D01, set the trigger coupling to dc, the trigger source to external, and the trigger level and slope to positive for test conditions K1-K10 and K21-K30. The trigger slope shall be set to negative for test conditions K11-K20 and K31-K40.
   c. Adjust output of item H01 for a frequency of 9.0 Hz ±10% and an amplitude of 2.5 ±0.1 Vp as read on item D01, channel B.
   d. The contact bounce shall be read on item D01, channel A and shall be as specified for test condition K1.
   e. Repeat notes a through d for test conditions K2-K40.

FIGURE 9. Operate time, release time, and contact bounce test circuit for tables IX, X, and XI - Continued.

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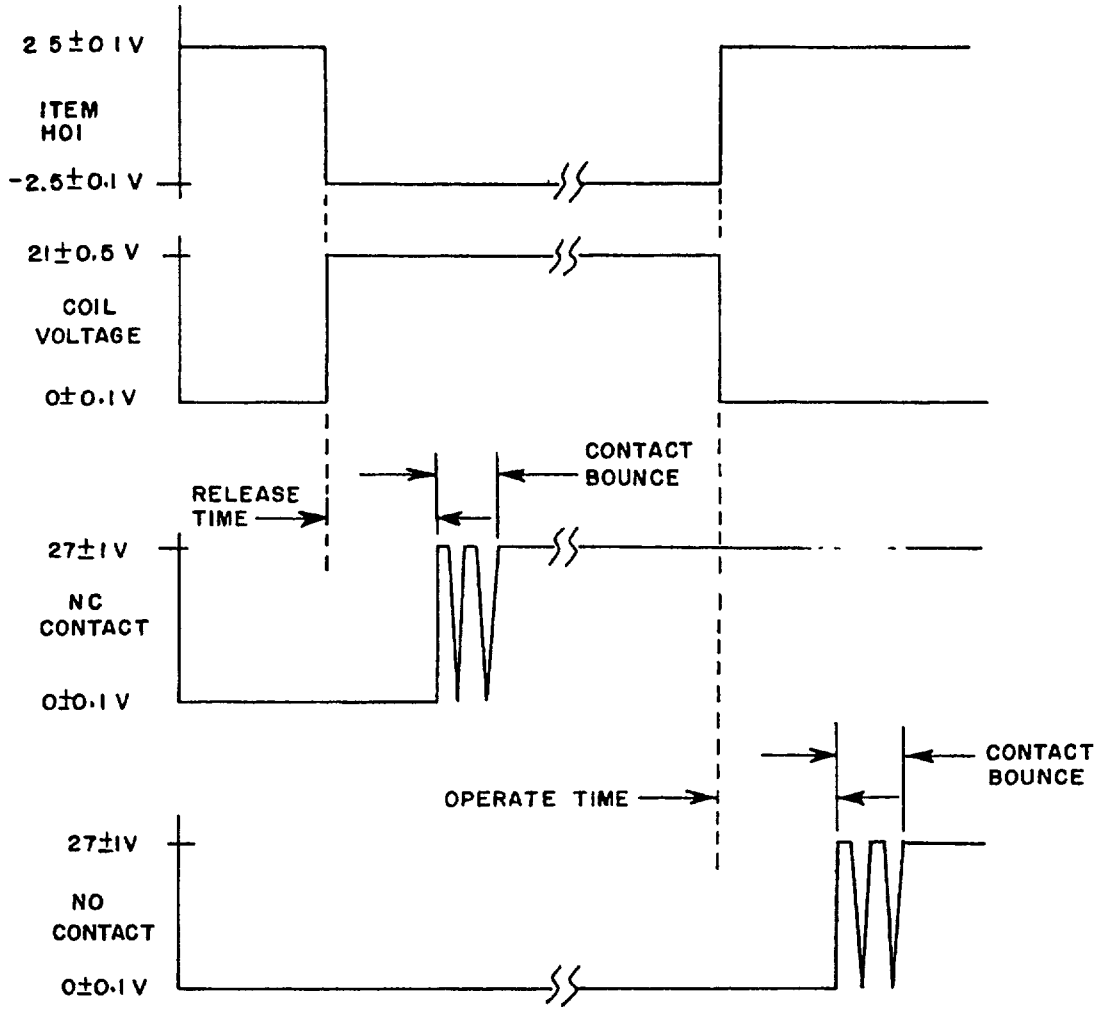


FIGURE 10. Typical oscilloscope traces used to determine the operate time, release time, and contact bounce for tables IX, X, and XI.

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